Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20220906-033-DFR-2	Issued	2022-09-16

Daily Field Report 2 - UT Thickness Survey - ATU-100 Sour Gas, Regenerator Overhead, Rich Amine

Scope and Responsibilities

Marcus Chen, acting as lead UT technician, performed ultrasonic thickness measurements on three lines within the ATU-100. The scope of work involved the execution of a UT thickness survey to monitor asset integrity for rich amine service.

All activities were conducted under the requirements of QSCP-NDT-001 Rev 2. The technician ensured that all field operations adhered to the safety protocols established in LNDE-JSA-20220906-033 and the authorizations granted via QSCP-WP-2022-09151.

Equipment and Calibration

Thickness measurements were performed using an Olympus 38DL PLUS 2847193 and an Olympus 38DL PLUS 2901456. Instrument verification and calibration were confirmed against LNDE-CAL-2022-0791 and LNDE-CAL-2021-0754.

Calibration was performed using LNDE-CS-001, a 4-step carbon steel calibration block ranging from 0.100-0.400 in. All measurements were conducted in accordance with the standard ASTM E797.

Inspection of Line 4-RA-102

The survey was performed on line 4-RA-102, carrying rich amine. This A106 Gr.B line has a nominal thickness of 8.56 mm and an outside diameter of 114.3 mm.

The measurements were evaluated against a t-min of 3.81 mm. The design limits for this section are 862 kPa(g) and 82 °C.

Inspection of Line 6-RA-103

Thickness data was collected for the rich amine header, line 6-RA-103. The line is constructed from A106 Gr.B with an outside diameter of 168.3 mm and a nominal thickness of 10.97 mm.

The retirement thickness for this line is 4.57 mm. Design parameters are specified as 862 kPa(g) and 82 °C.

Inspection of Line 4-RA-104

The final survey area covered the L/R exchanger rich side, line 4-RA-104. This A106 Gr.B line features an outside diameter of 114.3 mm and a nominal thickness of 8.56 mm.

The minimum allowable thickness is 3.81 mm. The design limits for this line are 862 kPa(g) and 127 °C.

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2022-09-16	_____